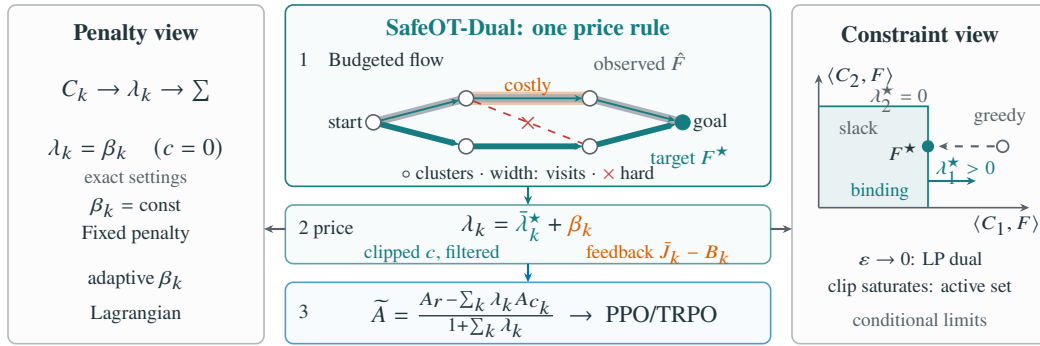
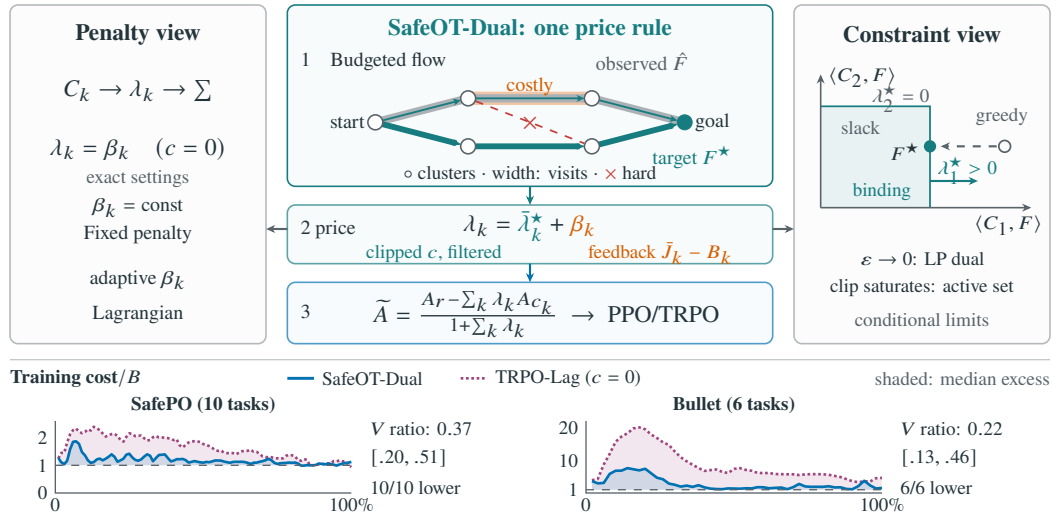


Print proof — 5.5-inch figure width

Body-text reference: 10 pt. Print at 100%; do not fit or shrink.



One price rule connects penalty and constraint views. The graph redirects observed visitation (gray) toward a budgeted target (teal); edge width encodes visits and crosses mark removed hard edges. Clipped graph prices combine with realized-cost feedback before the actor update. Zero graph-price cap gives exact penalty settings. LP and saturated-clip limits require the assumptions in Section 4; the target flow is not executed.



Graph-derived prices redirect learning while one rule connects penalty and constraint views. Observed flow is gray; the target is teal and is not executed. Hard edges are removed. LP and saturated-clip limits are conditional (Section 4). The strip shades median training-cost excess; V ratios report primary summaries: SafePO’s five-seed task median and Bullet’s ratio of means, with 95% intervals (Appendix).

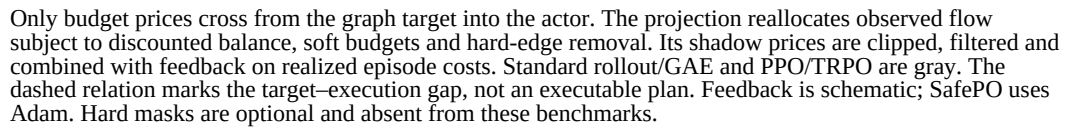
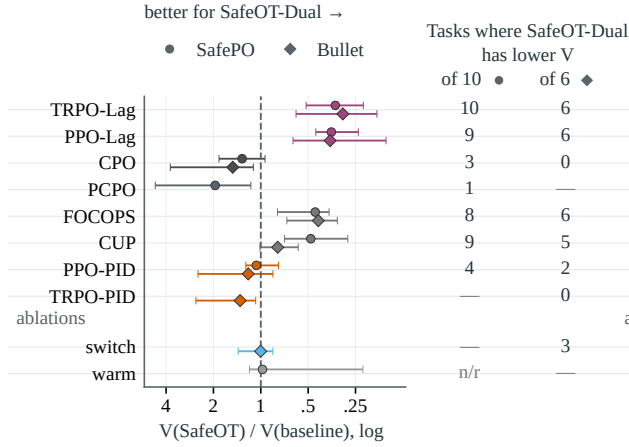
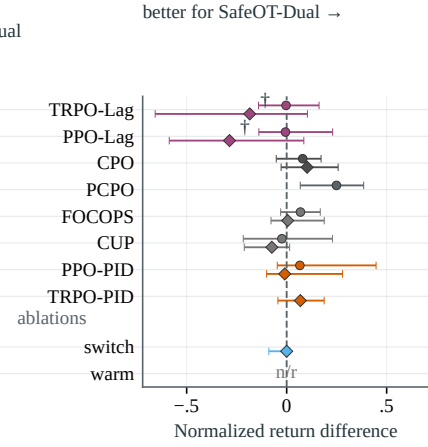


Figure2 · layout proof only

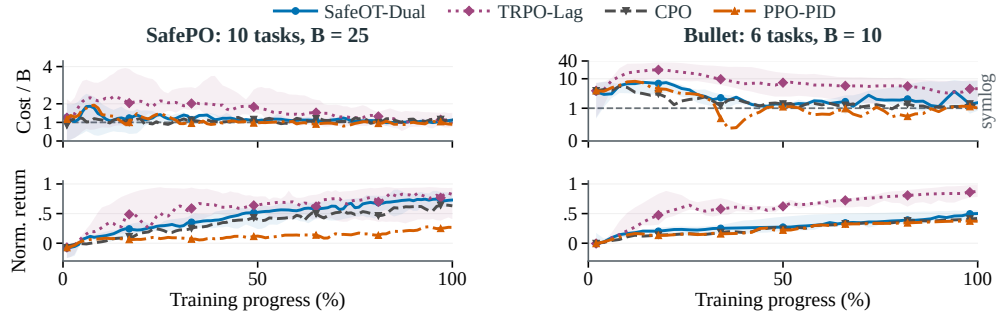
(a) Learning-time violation



(b) Final return



(c) All-task learning curves



Solved prices reduce learning-time violation versus Lagrangian baselines across both suites. (a,b) Right favors SafeOT-Dual: task-median V ratios (reversed log axis) and normalized final-return differences, with 95% bootstrap intervals. Circles denote SafePO; diamonds, Bullet. Counts show tasks with lower V. Dashes mean not run in that suite; n/r means not reported. † Final task-mean cost exceeds B on 4/6 TRPO-Lag and 5/6 PPO-Lag tasks. (c) Lines are task medians of seed means; blue/purple bands show task IQR. Budget equals 1; Bullet cost uses symlog. Normalizers, seed coverage and per-task curves are in the appendix.